

PRACTICA 2 : CONFIGURACIÓN AVANZADA DE SERVIDORES WEB Y HTTPS



Parte 1: Servidor Apache con PHP (Puerto 8080)

Se instala y configura el servidor Apache para servir contenido PHP en el puerto 8080.

Actualiza la lista de paquetes y mejora el sistema a las últimas versiones.

[illegible]

Instalación apache2

```
root@kali:~# apt-get install apache2
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
apache2 is already the newest version (2.4.58-1ubuntu9.9).
The following package was automatically installed and is no longer required:
  liblvm19
Use 'dpkg --get-selections' to remove it.
0 upgraded, 0 newly installed, 0 to remove and 12 not upgraded.
root@kali:~# apt-get install apache2
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
apache2 is already the newest version (2.4.58-1ubuntu9.9).
The following package was automatically installed and is no longer required:
  liblvm19
Use 'dpkg --get-selections' to remove it.
0 upgraded, 0 newly installed, 0 to remove and 12 not upgraded.
```

Abre el archivo de configuración de puertos. Cambia Listen 80 por Listen 8080

```
rodrigo@A6Alumno11:/mnt/c/WINDOWS/system32$ sudo nano /etc/apache2/ports.conf
```

[illegible]

Modificar el VirtualHost

[illegible]

Instalar PHP

```
rodrigo@A6Alumno11:/mnt/c/WINDOWS/system32$ sudo apt install php libapache2-mod-php -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
php is already the newest version (2:8.3+93ubuntu2).
libapache2-mod-php is already the newest version (2:8.3+93ubuntu2).
The following package was automatically installed and is no longer required:
  liblvm29
Use 'sudo apt autoremove' to remove it.
0 upgraded, 0 newly installed, 0 to remove and 12 not upgraded.
```

Verificar estado de Apache y Reiniciar Apache

```
rodrigo@A6Alumno11:/mnt/c/WINDOWS/system32$ sudo systemctl restart apache2
rodrigo@A6Alumno11:/mnt/c/WINDOWS/system32$ sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
   Active: active (running) since Thu 2025-10-09 14:32:33 CEST; 7s ago
     Docs: https://httpd.apache.org/docs/2.4/
   Process: 1516 ExecStart=/usr/sbin/apachectl start (code=exited, status=0/SUCCESS)
   Main PID: 1519 (apache2)
    Tasks: 6 (limit: 9350)
   Memory: 13.0M (peak: 14.0M)
      CPU: 43ms
   CGroup: /system.slice/apache2.service
           └─1519 /usr/sbin/apache2 -k start
             └─1521 /usr/sbin/apache2 -k start
               └─1522 /usr/sbin/apache2 -k start
                 └─1523 /usr/sbin/apache2 -k start
                   └─1524 /usr/sbin/apache2 -k start
                     └─1525 /usr/sbin/apache2 -k start

Oct 09 14:32:33 A6Alumno11 systemd[1]: Starting apache2.service - The Apache HTTP Server...
Oct 09 14:32:33 A6Alumno11 systemd[1]: Started apache2.service - The Apache HTTP Server.
rodrigo@A6Alumno11:/mnt/c/WINDOWS/system32$
```

Crear archivo PHP de prueba

```
rodrigo@A6Alumno11:/mnt/c/WINDOWS/system32$ echo "<?php phpinfo(); ?>" | sudo tee /var/www/html/info.php
<?php phpinfo(); ?>
```

Verifica que Apache sirve correctamente el contenido PHP



Parte 2: Servidor Nginx con HTML (Puerto 8081)

Se instala y configura el servidor NGINX para servir contenido HTML en el puerto 8081.

Instalo el servidor web Nginx

```
rodrimg723@Portatil-Rodrigo:/mnt/c/WINDOWS/system32$ sudo apt install nginx -y
[sudo] password for rodrimg723:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
nginx is already the newest version (1.24.0-2ubuntu7.5).
The following package was automatically installed and is no longer required:
  liblvm19
```

Creo una página HTML identificable para Nginx.

```
rodrimg723@Portatil-Rodrigo:/mnt/c/WINDOWS/system32$ echo "<h1>Servidor Nginx</h1><p>Funcionando en puer
to 8081</p>" | sudo tee /usr/share/nginx/html/index.html
<h1>Servidor Nginx</h1><p>Funcionando en puerto 8081</p>
```

Configuro Nginx en puerto 8081

```
GNU nano 7.2 /etc/nginx/sites-available/default
##
# You should look at the following URL's in order to grasp a solid understanding
# of Nginx configuration files in order to fully unleash the power of Nginx.
# https://www.nginx.com/resources/wiki/start/
# https://www.nginx.com/resources/wiki/start/topics/tutorials/config_pitfalls/
# https://wiki.debian.org/Nginx/DirectoryStructure
#
# In most cases, administrators will remove this file from sites-enabled/ and
# leave it as reference inside of sites-available where it will continue to be
# updated by the nginx packaging team.
#
# This file will automatically load configuration files provided by other
# applications, such as Drupal or Wordpress. These applications will be made
# available underneath a path with that package name, such as /drupal8.
#
# Please see /usr/share/doc/nginx-doc/examples/ for more detailed examples.
##

# Default server configuration
#
server {
    listen 8081 default_server;
    listen [::]:8081 default_server;

    # SSL configuration
    #
    # listen 443 ssl default_server;
    # listen [::]:443 ssl default_server;
    #
    # Note: You should disable gzip for SSL traffic.
    # See: https://bugs.debian.org/773332
    #
    # Read up on ssl_ciphers to ensure a secure configuration.
    # See: https://bugs.debian.org/765782
    #
    # Self signed certs generated by the ssl-cert package
    # Don't use them in a production server!
    #
    # include snippets/snakeoil.conf;

    root /var/www/html;

    # Add index.php to the list if you are using PHP
    index index.html index.htm index.nginx-debian.html;

    server_name _;

    location / {
        # First attempt to serve request as file, then
        # as directory, then fall back to displaying a 404.
        try_files $uri $uri/ =404;
    }

    # pass PHP scripts to FastCGI server
    #
    #location ~ \.php$ {
    #    include snippets/fastcgi-php.conf;
    #
    #
```

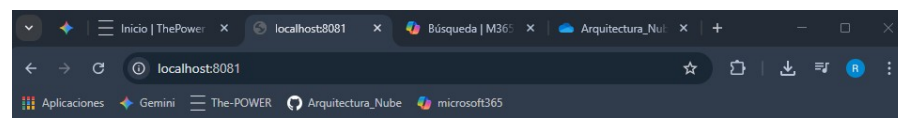
Reinicio Nginx para aplicar los cambios y Compruebo que Nginx está funcionando correctamente en el puerto 8081

```
rodrigo@A6Alumno11:/mnt/c/WINDOWS/system32$ sudo systemctl restart nginx
rodrigo@A6Alumno11:/mnt/c/WINDOWS/system32$ sudo systemctl status nginx
● nginx.service - A high performance web server and a reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
   Active: active (running) since Tue 2025-10-14 09:00:27 CEST; 10s ago
     Docs: man:nginx(8)
   Process: 1537 ExecStartPre=/usr/sbin/nginx -t -q -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
   Process: 1538 ExecStart=/usr/sbin/nginx -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
   Main PID: 1540 (nginx)
    Tasks: 17 (limit: 9350)
   Memory: 13.1M (peak: 15.0M)
      CPU: 41ms
   CGroup: /system.slice/nginx.service
           └─1540 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;"
             └─1541 "nginx: worker process"
               └─1542 "nginx: worker process"
                 └─1543 "nginx: worker process"
                   └─1544 "nginx: worker process"
                     └─1545 "nginx: worker process"
                       └─1546 "nginx: worker process"
                         └─1547 "nginx: worker process"
                           └─1548 "nginx: worker process"
                             └─1550 "nginx: worker process"
                               └─1551 "nginx: worker process"
                                 └─1552 "nginx: worker process"
                                   └─1553 "nginx: worker process"
                                     └─1554 "nginx: worker process"
                                       └─1555 "nginx: worker process"
                                         └─1556 "nginx: worker process"
                                           └─1557 "nginx: worker process"

Oct 14 09:00:27 A6Alumno11 systemd[1]: Starting nginx.service - A high performance web server and a reverse proxy se
Oct 14 09:00:27 A6Alumno11 systemd[1]: Started nginx.service - A high performance web server and a reverse proxy ser
lines 1-31/31 (END)
```

Verificación de que Nginx sirve correctamente el contenido HTML

```
rodrigo@A6Alumno11:/mnt/c/WINDOWS/system32$ curl http://localhost:8081
<h1>Hola Mundo desde Nginx</h1><p>Servidor funcionando correctamente</p>
```



Hola Mundo desde Nginx

Servidor funcionando correctamente

PARTE 3: INSTALACIÓN Y CONFIGURACIÓN DE CADDY

Instalar dependencias necesarias

```
rodrigo@A6Alumno11:/mnt/c/WINDOWS/system32$ sudo apt install -y debian-keyring debian-archive-keyring apt-transport-https curl
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
apt-transport-https is already the newest version (2.8.3).
curl is already the newest version (8.5.0-2ubuntu10.6).
The following package was automatically installed and is no longer required:
  liblvm19
Use 'sudo apt autoremove' to remove it.
The following NEW packages will be installed:
  debian-archive-keyring debian-keyring
0 upgraded, 2 newly installed, 0 to remove and 13 not upgraded.
Need to get 31.5 MB of archives.
After this operation, 33.4 MB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble/universe amd64 debian-archive-keyring all 2023.4ubuntu1 [168 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble/universe amd64 debian-keyring all 2023.12.24 [31.3 MB]
Fetched 31.5 MB in 1s (49.9 MB/s)
Selecting previously unselected package debian-archive-keyring.
(Reading database ... 43474 files and directories currently installed.)
Preparing to unpack .../debian-archive-keyring_2023.4ubuntu1_all.deb ...
Unpacking debian-archive-keyring (2023.4ubuntu1) ...
Selecting previously unselected package debian-keyring.
Preparing to unpack .../debian-keyring_2023.12.24_all.deb ...
Unpacking debian-keyring (2023.12.24) ...
Setting up debian-archive-keyring (2023.4ubuntu1) ...
Setting up debian-keyring (2023.12.24) ...
```

Agregar repositorio de Caddy

```
rodrigo@A6Alumno11:/mnt/c/WINDOWS/system32$ curl -1sLf 'https://dl.cloudsmith.io/public/caddy/stable/gpg.key' | sudo
gpg --dearmor -o /usr/share/keyrings/caddy-stable-archive-keyring.gpg
rodrigo@A6Alumno11:/mnt/c/WINDOWS/system32$ curl -1sLf 'https://dl.cloudsmith.io/public/caddy/stable/debian.deb.txt'
| sudo tee /etc/apt/sources.list.d/caddy-stable.list
# Source: Caddy
# Site: https://github.com/caddyserver/caddy
# Repository: Caddy / stable
# Description: Fast, multi-platform web server with automatic HTTPS

deb [signed-by=/usr/share/keyrings/caddy-stable-archive-keyring.gpg] https://dl.cloudsmith.io/public/caddy/stable/deb
/debian any-version main

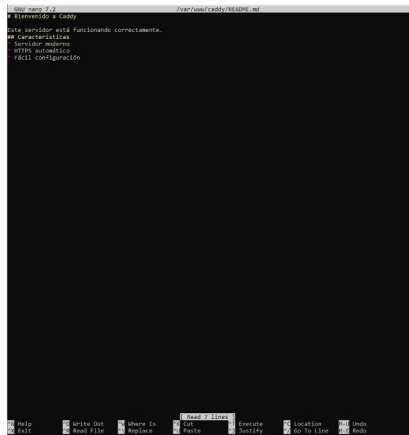
deb-src [signed-by=/usr/share/keyrings/caddy-stable-archive-keyring.gpg] https://dl.cloudsmith.io/public/caddy/stable
/deb/debian any-version main
```

Actualizar e instalar Caddy

```
rodrigo@A6Alumno11:/mnt/c/WINDOWS/system32$ sudo apt update && sudo apt install caddy -y
Get:1 https://dl.cloudsmith.io/public/caddy/stable/deb/debian any-version InRelease [14.8 kB]
Get:2 https://download.docker.com/linux/ubuntu noble InRelease [48.5 kB]
Hit:3 http://security.ubuntu.com/ubuntu noble-security InRelease
Err:2 https://download.docker.com/linux/ubuntu noble InRelease
  The following signatures couldn't be verified because the public key is not available: NO_PUBKEY 7EA0A9C3F273FCD8
Get:4 https://dl.cloudsmith.io/public/caddy/stable/deb/debian any-version/main amd64 Packages [4329 B]
Hit:5 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:6 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:7 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Fetched 67.6 kB in 0s (146 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
13 packages can be upgraded. Run 'apt list --upgradable' to see them.
W: An error occurred during the signature verification. The repository is not updated and the previous index files will be used. GPG error: https://download.docker.com/linux/ubuntu noble InRelease: The following signatures couldn't be verified because the public key is not available: NO_PUBKEY 7EA0A9C3F273FCD8
W: Failed to fetch https://download.docker.com/linux/ubuntu/dists/noble/InRelease The following signatures couldn't be verified because the public key is not available: NO_PUBKEY 7EA0A9C3F273FCD8
W: Some index files failed to download. They have been ignored, or old ones used instead.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following package was automatically installed and is no longer required:
  liblvm19
Use 'sudo apt autoremove' to remove it.
The following NEW packages will be installed:
  caddy
0 upgraded, 1 newly installed, 0 to remove and 13 not upgraded.
Need to get 16.0 MB of archives.
After this operation, 44.5 MB of additional disk space will be used.
Get:1 https://dl.cloudsmith.io/public/caddy/stable/deb/debian any-version/main amd64 caddy amd64 2.10.2 [16.0 MB]
Fetched 16.0 MB in 1s (28.4 MB/s)
Selecting previously unselected package caddy.
(Reading database ... 43506 files and directories currently installed.)
Preparing to unpack .../caddy_2.10.2_amd64.deb ...
Unpacking caddy (2.10.2) ...
Setting up caddy (2.10.2) ...
Created symlink /etc/systemd/system/multi-user.target.wants/caddy.service → /usr/lib/systemd/system/caddy.service.
Processing triggers for man-db (2.12.0-4build2) ...
```

Crear directorio para Caddy y crear archivo Markdown de prueba

```
rodrigo@A6Alumno11:/mnt/c/WINDOWS/system32$ sudo mkdir -p /var/www/caddy
rodrigo@A6Alumno11:/mnt/c/WINDOWS/system32$ echo "# Bienvenido a Caddy" | sudo tee /var/www/caddy/README.md
# Bienvenido a Caddy
rodrigo@A6Alumno11:/mnt/c/WINDOWS/system32$ echo "## Características" | sudo tee -a /var/www/caddy/README.md
## Características
rodrigo@A6Alumno11:/mnt/c/WINDOWS/system32$ echo "* Servidor moderno" | sudo tee -a /var/www/caddy/README.md
* Servidor moderno
rodrigo@A6Alumno11:/mnt/c/WINDOWS/system32$ echo "* HTTPS automático" | sudo tee -a /var/www/caddy/README.md
* HTTPS automático
rodrigo@A6Alumno11:/mnt/c/WINDOWS/system32$ echo "* Fácil configuración" | sudo tee -a /var/www/caddy/README.md
* Fácil configuración
rodrigo@A6Alumno11:/mnt/c/WINDOWS/system32$ sudo nano /var/www/caddy/README.md
```

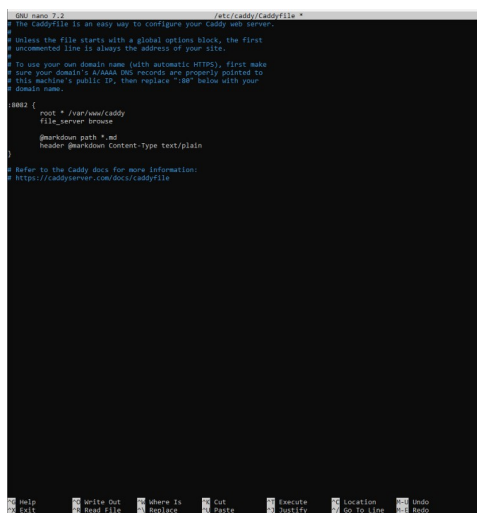


Creo una imagen de prueba ajustado a wsl

```
rodrigo@A6Alumno11:/mnt/c/WINDOWS/system32$ curl -o /tmp/test-image.jpg https://www.python.org/static/apple-touchicon-144x144-precomposed.png
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left   Speed
100 146 100 146    0     0  912      0 --:--:-- --:--:-- --:--:--  918
rodrigo@A6Alumno11:/mnt/c/WINDOWS/system32$ sudo mv /tmp/test-image.jpg /var/www/caddy/test.jpg
rodrigo@A6Alumno11:/mnt/c/WINDOWS/system32$
```

Crear Caddyfile personalizado con información sobre el puerto y más información

```
rodrigo@A6Alumno11:/mnt/c/WINDOWS/system32$ sudo nano /etc/caddy/Caddyfile
```



Reinicio de Caddy y verificación del estado de Caddy

```
rodring723@Portatil-Rodrigo:/mnt/c/WINDOWS/system32$ sudo systemctl restart caddy
```

```

# cd /etc/nginx/sites-enabled; sudo systemctl status caddy
caddy.service - Caddy
Loaded: loaded (/usr/lib/systemd/system/caddy.service; enabled; preset: enabled)
Active: active (running) since Tue 2025-10-14 09:20:24 CEST; 6s ago
Docs: https://caddyserver.com/docs/
Main PID: 2453 (caddy)
Tasks: 11 (limit 3938)
Memory: 11.5M (peak: 12.6M)
CPU: 42ms
Group: system.slice/caddy.service
└─3453 /usr/bin/caddy run --envion --config /etc/caddy/Caddyfile

Oct 14 09:20:24 AGA1uom01 caddy[2453]: ["level": "info", "ts": "1760426344.415779", "logger": "admin", "msg": "admin endpoint
Oct 14 09:20:24 AGA1uom01 caddy[2453]: ["level": "warn", "ts": "1760426424.441869", "logger": "http", "msg": "HTTP/2 skipped
Oct 14 09:20:24 AGA1uom01 caddy[2453]: ["level": "warn", "ts": "1760426424.441898", "logger": "http", "msg": "HTTP/3 skipped
Oct 14 09:20:24 AGA1uom01 caddy[2453]: ["level": "info", "ts": "1760426424.441898", "logger": "http.log", "msg": "server runn
Oct 14 09:20:24 AGA1uom01 caddy[2453]: ["level": "info", "ts": "1760426424.441875", "logger": "http.cache_maintenance", "ms
Oct 14 09:20:24 AGA1uom01 caddy[2453]: ["level": "info", "ts": "1760426424.442105", "msg": "autosaved config (load with --
Oct 14 09:20:24 AGA1uom01 caddy[2453]: ["level": "info", "ts": "1760426424.442171", "msg": "starting initial configuration"
Oct 14 09:20:24 AGA1uom01 caddy[2453]: ["level": "info", "ts": "1760426424.444600", "logger": "tls", "msg": "storage clean
Oct 14 09:20:24 AGA1uom01 caddy[2453]: ["level": "info", "ts": "1760426424.444665", "logger": "tls", "msg": "finished clean

```

Prueba de Caddy desde terminal y desde navegador

```
[root@0604080 ~]# npm run build && curl http://localhost:8082/
```

```
<!DOCTYPE html>  
<html>  
  <head>  
    <title></title>  
    <link rel="canonical" href="/" />  
    <meta charset="utf-8">  
    <meta name="color-scheme" content="light dark">  
    <meta name="viewport" content="width=device-width, initial-scale=1.0">  
  <style nonce="be9d0c_5f50_4f4a_a0d0_d4a3855896">  
    * { padding: 0; margin: 0; box-sizing: border-box; }  
  
body {  
  font-family: Inter, system-ui, sans-serif;  
  font-size: 16px;  
  text-rendering: optimizeSpeed;  
  background-color: #f3f6ff;  
  min-height: 100vh;  
  
img,  
svg {  
  vertical-align: middle;  
  z-index: 1;  
  
img {  
  max-width: 100%;  
  max-height: 100%;  
  border-radius: 5px;  
  
td img {  
  max-width: 1.5em;  
  max-height: 2em;  
  object-fit: cover;  
  
body,  
svg,  
layout-current,  
layout-current svg,  
go-up {  
  color: #333;  
  text-decoration: none;  
  
} layout-list, layout-grid {  
  cursor: pointer;  
  
} wrapper {  
  max-width: 1200px;  
  margin-left: auto;  
  margin-right: auto;  
  
header,
```

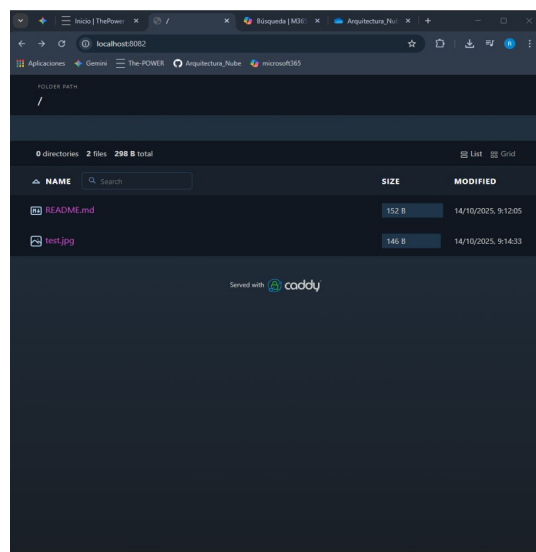
```

Rodrigo@A6A1ummo11: /mnt/c/WINDOWS/system32$ curl http://localhost:8082/README.md
# Bienvenido a Caddy

Este servidor está funcionando correctamente.

## Características
* Servidor moderno
* HTTPS automático
* Fácil configuración

```



Instalación de Certbot y plugin de Apache

Creación de un certificado autofirmado para practicar HTTPS localmente rellenando los datos necesarios

Activación del módulo SSL necesario para HTTPS en Apache

Creación configuración SSL para Apache y modificación de VirtualHost SSL

```

podriga@A6A1umno11:/mnt/c/WINDOWS/system32$ sudo nano /etc/apache2/sites-available/default-ssl.conf
(VirtualHost *:8443)
#DocumentRoot /var/www/html

#
# Available loglevels: trace, ..., trace2, debug, info, notice, warn,
# error, crit, alert, emerg.
# It is also possible to configure the loglevel for particular
# modules, e.g.
#LogLevel info ssl:warn

ErrorLog ${APACHE_LOG_DIR}/error.log
CustomLog ${APACHE_LOG_DIR}/access.log combined

#
# For most configuration files from conf-available, which are
# enabled or disabled at a global level, it is possible to
# include a line for only one particular virtual host. For example the
# following line enables the CGI configuration for this host only
# after it has been globally disabled with "addiscon".
#Include conf-available/serve-cgi-bin.conf

# SSL Engine Switch:
# Enable/disable SSL for this virtual host.
SSLEngine on

#
# A self-signed (snakeoil) certificate can be created by installing
# the ssl-cert package. See
# /usr/share/doc/openssl/README.Debian.gz for more info.
# If both key and certificate are stored in the same file, only the
# SSLCertificate directive is needed.
SSLCertificateFile /etc/ssl/certs/apache-selfsigned.crt
SSLCertificateKeyFile /etc/ssl/private/apache-selfsigned.key

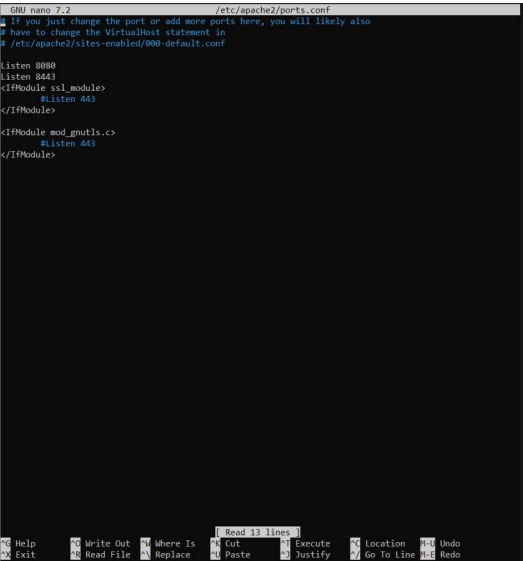
#
# Server Certificate Chain:
# Point to a chain of subcertificates in a file containing the
# concatenation of PEM encoded CA certificates which form the
# certificate chain for the server certificate. Alternatively
# the referenced file can be the same as SSLCertificateFile
# when the CA certificates are directly appended to the server
# certificate for convenience.
#SSLCertificateChainFile /etc/apache2/ssl.crt/server-ca.crt

#
# Certificate Authority (CA):
# Set the CA certificate verification path where to find CA
# certificates for client authentication or alternatively one
# huge file containing all of them (file must be PEM encoded)
# Note: Inside SSLCertificatePath you need hash symlinks
# to point to the certificate files. Use the provided
# Makefile to update the hash symlinks after changes.
#SSLCACertificatePath /etc/ssl/certs/
SSLCACertificateFile /etc/apache2/ssl.crt/ca-bundle.crt

#
# Certificate Revocation Lists (CRL):
# Set the CA revocation path where to find CA CRLs for client
# authentication or alternatively one huge file containing all
# of them (file must be PEM encoded)
# Note: Inside SSLCRLocation you need hash symlinks
# to point to the certificate files. Use the provided

```

Cambio del puerto SSL al puerto 8443



Habilitamos sitio SSL con el siguiente comando

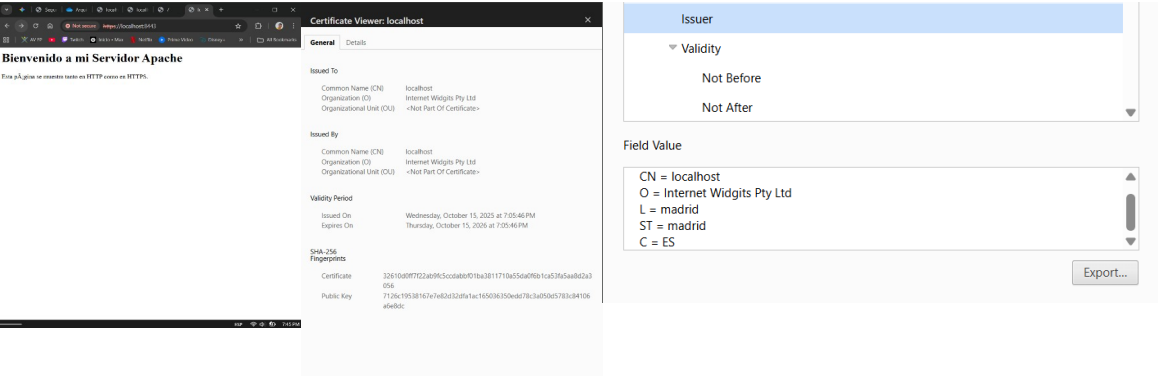
```
rodrimg723@Portatil-Rodrigo:/mnt/c/WINDOWS/system32$ sudo a2ensite default-ssl.conf
Enabling site default-ssl.
To activate the new configuration, you need to run:
systemctl reload apache2
```

Después de reiniciar apache verificamos en terminal y en navegador el correcto funcionamiento

```
rodrimg723@Portatil-Rodrigo:/mnt/c/WINDOWS/system32$ curl -i -k https://localhost:8443
HTTP/1.1 200 OK
Date: Sat, 18 Oct 2025 10:01:09 GMT
Server: Apache/2.4.58 (Ubuntu)
Last-Modified: Wed, 15 Oct 2025 17:13:22 GMT
ETag: "64-641359e1e0bdf"
Accept-Ranges: bytes
Content-Length: 100
Vary: Accept-Encoding
Content-Type: text/html

<h1>Bienvenido a mi Servidor Apache</h1><p>Esta página se muestra tanto en HTTP como en HTTPS.</p>
```

También muestro la verificación de que el certificado es el que yo cree en los pasos anteriores



PART 5: VERIFICACIÓN FINAL DE LOS TRES SERVIDORES

Comprobación del estado activo de los tres servidores simultáneamente

```
rodrimg723@Portatil-Rodrigo:/mnt/c/WINDOWS/system32$ sudo systemctl status apache2 nginx caddy
[sudo] password for rodrimg723:
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
   Active: active (running) since Wed 2025-10-15 19:13:34 CEST; 18min ago
     Docs: https://httpd.apache.org/docs/2.4/
   Process: 12999 ExecStart=/usr/sbin/apachectl start (code=exited, status=0/SUCCESS)
  Main PID: 13002 (apache2)
    Tasks: 7 (limit: 4315)
   Memory: 13.4M (peak: 14.5M)
      CPU: 351ms
   CGroup: /system.slice/apache2.service
           └─13002 /usr/sbin/apache2 -k start
             └─13004 /usr/sbin/apache2 -k start
               └─13005 /usr/sbin/apache2 -k start
                 └─13006 /usr/sbin/apache2 -k start
                   └─13007 /usr/sbin/apache2 -k start
                     └─13008 /usr/sbin/apache2 -k start
                       └─13012 /usr/sbin/apache2 -k start

Oct 15 19:13:33 Portatil-Rodrigo systemd[1]: Starting apache2.service - The Apache HTTP Server...
Oct 15 19:13:34 Portatil-Rodrigo systemd[1]: Started apache2.service - The Apache HTTP Server.

● nginx.service - A high performance web server and a reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
   Active: active (running) since Wed 2025-10-15 18:19:03 CEST; 1h 12min ago
     Docs: man:nginx(8)
  Main PID: 10813 (nginx)
    Tasks: 9 (limit: 4315)
   Memory: 6.4M (peak: 7.5M)
      CPU: 96ms
   CGroup: /system.slice/nginx.service
           └─10813 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;"
             └─10814 "nginx: worker process"
               └─10815 "nginx: worker process"
                 └─10816 "nginx: worker process"
                   └─10817 "nginx: worker process"
                     └─10818 "nginx: worker process"
                       └─10819 "nginx: worker process"
                         └─10820 "nginx: worker process"
                           └─10822 "nginx: worker process"

● caddy.service - Caddy
   Loaded: loaded (/usr/lib/systemd/system/caddy.service; enabled; preset: enabled)
   Active: active (running) since Wed 2025-10-15 18:32:29 CEST; 59min ago
     Docs: https://caddyserver.com/docs/
  Main PID: 12107 (caddy)
    Tasks: 12 (limit: 4315)
   log file: ─
```

Listado de los puertos donde están escuchando los servidores

```
rodrimg723@Portatil-Rodrigo:/mnt/c/WINDOWS/system32$ sudo netstat -tulpn | grep -E '8080|8081|8082|8443'
tcp        0      0 0.0.0.0:8081          0.0.0.0:*           LISTEN     10813/nginx: master
tcp6       0      0 :::8080              :::*                 LISTEN     13002/apache2
tcp6       0      0 :::8082              :::*                 LISTEN     12107/caddy
tcp6       0      0 :::8443              :::*                 LISTEN     13002/apache2
```

```

root@img723@Portatil-Rodrigo:/mnt/c/WINDOWS/system32$ curl http://localhost:8080
<h1>Bienvenido a mi Servidor Apache</h1><p>Esta página se muestra tanto en HTTP como en HTTPS.</p>
root@img723@Portatil-Rodrigo:/mnt/c/WINDOWS/system32$ curl http://localhost:8081
<h1>Servidor Nginx</h1><p>Funcionando en puerto 8081</p>
root@img723@Portatil-Rodrigo:/mnt/c/WINDOWS/system32$ curl http://localhost:8082

<!DOCTYPE html>
<html>

  <head>

    <title></title>
    <link rel="canonical" href="/" />
    <meta charset="utf-8">
    <meta name="color-scheme" content="light dark">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">

<style nonce="593da304-a31b-43c3-b635-9ebacf88d899">
* { padding: 0; margin: 0; box-sizing: border-box; }

body {
  font-family: Inter, system-ui, sans-serif;
  font-size: 16px;
  text-rendering: optimizeSpeed;
  background-color: #f3f6f7;
  min-height: 100vh;
}

img,
svg {
  vertical-align: middle;
  z-index: 1;
}

img {
  max-width: 100%;

```

